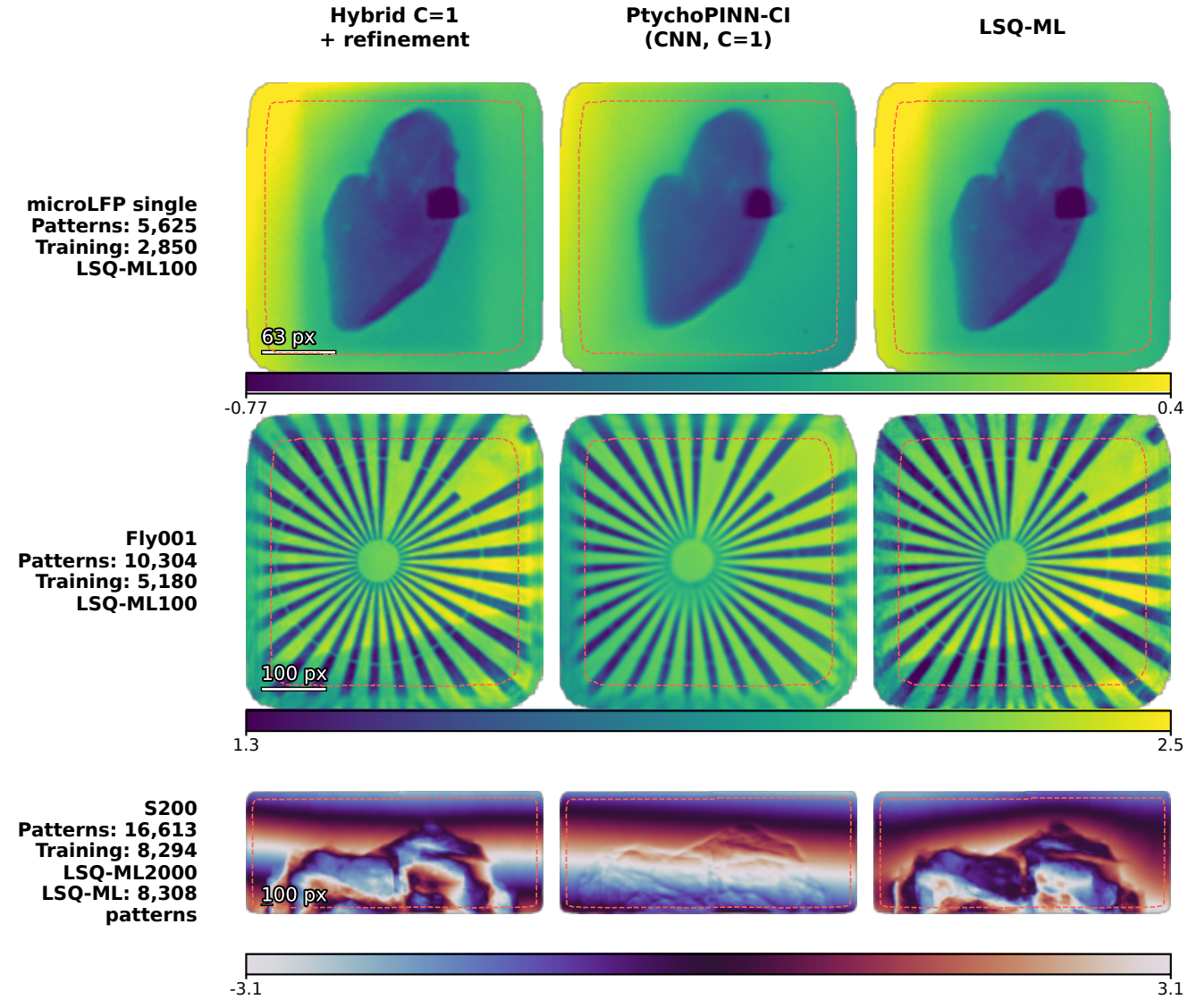



Phase is shown in radians; amplitude shows relative object amplitude. Pattern counts refer to the displayed acquisition and distinct training patterns. High half selects checkpoints (spatial validation, not an independent test; no guard band). Reference is an experimental estimate; dashed contour marks scored interior. Locally trained PtychoPINN-CI (CNN, C=1) uses the same measured training half as Hybrid ResNet. No patch-offset alignment, tilt seeding or phase-field refinement is applied; intrinsic CI calibration and scalar count calibration are retained. This is a local measured-data result, distinct from the published synthetic-only reconstruction. Whole-acquisition images include the training region. Paired groups assign C4 only where it has lower phase MAE after refinement and higher phase SSIM; other datasets use C1. This is the C1-selected group. S200 LSQ-ML2000: historical 8,308-pattern subset versus 16,613 neural patterns; iteration-budget limited, not a matched timing result.